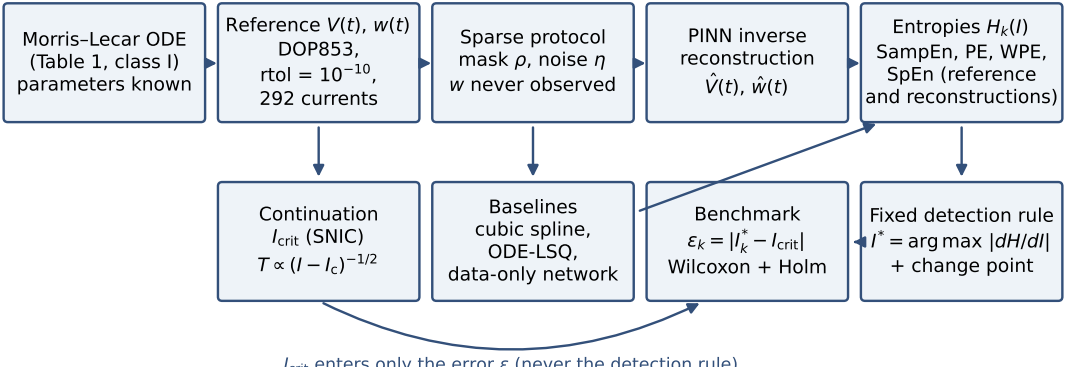
Morris–Lecar ODE (Table 1, class I) parameters known
Reference $V(t)$, $w(t)$ DOP853, rtol = 10^{-10}, 292 currents
Sparse protocol mask ρ, noise η w never observed
PINN inverse reconstruction $\hat{V}(t)$, $\hat{w}(t)$
Entropies $H_k(I)$ SampEn, PE, WPE, SpEn (reference and reconstructions)
Continuation I_{crit} (SNIC) $T \propto (I - I_c)^{-1/2}$
Baselines cubic spline, ODE-LSQ, data-only network
Benchmark $\varepsilon_k = |I_k^* - I_{crit}|$ Wilcoxon + Holm
Fixed detection rule $I^* = \arg\max |dH/dI|$ + change point
I_{crit} enters only the error ε (never the detection rule)